Exam 2 - Conceptual

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Question 1 pts

A regional airline's quality manager has implemented two new initiatives: supervisors now conduct random in-flight service audits each week, and the maintenance team performs detailed equipment inspections on every aircraft before departure.

These initiatives are examples of:

- prevention costs
- internal failure costs
- external failure costs
- appraisal costs

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Question 1 pts

The application of statistical techniques to determine whether a quantity of material should be accepted or rejected based on the inspection or test of a sample is known as

- specification review
- the Deming Wheel
- benchmarking
- acceptance sampling

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Question 1 pts

At Imroze Bakery, the operations manager has been conducting waste walks across the production floor.
Match each observed situation to the correct type of waste.
Baked loaves are held in large staging bins for hours before packaging begins, tying up floor space and making batch tracking difficult.

Inventory

Bakers frequently walk to the far end of the kitchen to retrieve ingredient bins that are not stored at their workstation.

Motion
The packaging line can wrap 200 loaves per hour, but the oven only outputs 120 loaves per hour, leaving packaging staff idle.

Waiting
Roughly 9% of loaves are rejected after cooling due to underbaking and must be discarded or repurposed, consuming labor and ingredients.

Defects
Other Incorrect Match Options:

- Transportation

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Question 1 pts

VineRidge Winery has been producing premium California wines for over two decades. When two of its largest rivals — Sonoma Hills and Pacific Crest Vineyards — each announced plans to expand their bottling and storage facilities by 35%, VineRidge's board quickly approved a comparable expansion, despite internal forecasts showing current capacity is still sufficient for the next two years.

Which capacity timing strategy is VineRidge following?

- Expansionist
- Follow the leader
- Wait-and-see
- Theory of Constraints

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Question 1 pts

In a DBR system, the mechanism that controls the rate at which the bottleneck dictates the throughput of the entire plant by syncing with the materials release schedule is called the:

- drum
- buffer
- kanban
- rope

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Question 1 pts

A paint shop tracks the number of surface defects per car body as it comes off the line. Each day, one car body is fully inspected, and the total number of surface defects is counted.

Which control chart should be used, and why?

- $\bar{X}$ and R chart — because paint quality is a continuous measurement taken in groups
- p-chart — because each car body either passes or fails inspection
- c-chart — because we are counting the number of defects on a single inspected unit, and one unit can have multiple defects
- p-chart — because sample sizes at automotive plants are always large

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Question 1 pts

A bicycle manufacturing company, Hikeroo, is implementing a comprehensive quality initiative. The operations manager presents the following action plan to the executive team:

Phase 1: Invest $2 million in state-of-the-art automated inspection equipment with AI-powered defect detection
Phase 2: Implement Six Sigma methodology with rigorous statistical process control across all production lines
Phase 3: Establish a centralized Quality Assurance Department with 15 dedicated specialists reporting directly to the VP of Operations
Phase 4: Require all suppliers to achieve ISO 9001 certification within 12 months

The manager concludes: *"This comprehensive plan addresses quality from every angle—technology, methodology, organizational structure, and supply chain. This represents a true Total Quality Management approach."*

Which statement best identifies the gap in this quality initiative relative to core TQM principles?

- The plan lacks sufficient investment in quality control technology and automation
- The plan focuses on inspection and control systems but doesn't emphasize customer satisfaction, continuous improvement culture, and company-wide employee involvement
- Six Sigma and ISO 9001 are incompatible with TQM and should not be included together
- The plan should establish quality as the sole responsibility of the Quality Assurance Department rather than distributing it across functions

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Question 1 pts

Harvest Fresh, a large-scale food packaging company, has been struggling with throughput limitations on its canning line.

The newly hired operations director, Marcus Webb, presents his improvement roadmap to the executive team: *"Here is our plan. First, we identify which station on the canning line is limiting our output. Second, we squeeze every unit of capacity we can out of that station without new investment. Third, we align everything else in the plant to support that station's pace. Fourth, if it's still the constraint after all that, we invest in additional capacity to break the limit. This is our TOC implementation plan."*

Marcus's plan represents the complete TOC methodology.

- True
- False

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Question 1 pts

CrestBrew, a regional craft beer company, has grown rapidly and is now evaluating whether to build a single large 120,000 sq ft brewing facility or stick with its current two smaller 60,000 sq ft facilities. The COO pushes back on the expansion plan, warning: "At that scale, diseconomies of scale may start to creep in — we should think carefully before going bigger."

What does the COO most likely mean by this warning?

- Doubling facility size doubles both fixed and variable costs, so average cost per unit stays constant — the expansion offers no real efficiency gain
- Beyond an optimal scale, average cost per unit begins to rise as coordination complexity, management overhead, and operational inefficiencies outweigh the benefits of size
- Concentrating all production in one large facility raises fixed costs significantly, so any drop in demand will sharply increase the cost per unit
- At that scale, the plant will require highly specialized brewing equipment, making it harder to adapt to new product lines or recipe changes

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Question 1 pts

A hospital emergency department sees a highly varied mix of patients — some need a quick prescription refill while others require complex trauma care. The department uses flexible staffing, allows considerable nurse discretion, and workflows differ patient to patient. A consultant recommends standardizing processes into fixed protocols to improve efficiency.

- Standardizing will increase capital intensity, which is always too expensive for service settings
- The recommendation would push the department off the strategic diagonal — imposing low-divergence structure on a high-contact, high-customization process
- Fixed protocols will increase resource flexibility, which conflicts with lean principles
- Standardization always leads to diseconomies of scale in service settings

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